Energy Futures Backtesting: A Simple EMA Strategy in Pine Script

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Objective and market

I developed and back tested a rule-based EMA crossover strategy on Micro WTI crude oil futures using Pine Script in TradingView. The project evaluates historical profitability and risk after accounting for commissions and simulated slippage. It also compares the baseline strategy with a 200-period EMA trend filter and examines performance over a separate evaluation period. The aim is to demonstrate a practical strategy-development and testing process.

Instrument: Micro WTI crude oil futures (NYMEX:MCL1!).
Timeframe: 2-hour candles.
Commission: $1.50 per contract.
Slippage: Two ticks per market-order fill.

Understanding Micro WTI Crude Oil Futures

Micro WTI Crude Oil futures (MCL) track the price of West Texas Intermediate crude oil. Each contract represents 100 barrels. For example, at an illustrative oil price of $70 per barrel, one contract provides $7,000 of market exposure. A 1% price move $0.70 per barrel therefore creates a $70 gain or loss per contract, before trading costs.

Oil prices are sensitive to political and economic developments. Wars, sanctions, and disruptions to major shipping routes can affect supply, while OPEC production decisions, inventory reports, and changes in global economic activity influence the supply-and-demand outlook. These events can cause sharp price movements, meaning that the smaller size of a Micro WTI contract reduces the dollar exposure but does not remove the underlying market risk.

Oil often has an inverse relationship with the US dollar. Because oil is priced in dollars, a stronger dollar makes it more expensive for buyers using other currencies, which can put pressure on demand and prices. A weaker dollar can provide support instead.

Strategy rules

Entry: Buy one futures contract when the 20-period EMA crosses above the 50-period EMA.
Exit: Close the entire position when the 20-period EMA crosses below the 50-period EMA.
Direction: Long only
Position size: One contract at a time; no additional entries while a position is open.
Risk management: No stop-loss or take-profit orders are used. Positions are closed solely on the downward EMA crossover.

Pinescript

V1| Simple crossover

|January 3, 2022 to September 18, 2025,|

```
strategy ("Energy Futures  EMA Backtest",
    overlay=true,
    initial_capital=10000,
    currency=currency.USD,
    default_qty_type=strategy.fixed,
    default_qty_value=1,
    pyramiding=0,
Prevents the strategy from adding further entries in the same direction while a position is already open.
    commission_type=strategy.commission.cash_per_contract,
charge a fixed cash commission for each contract traded.
    commission_value=1.50,
$1.50 per contract, per side.
    slippage=2,
simulated market-order fills two ticks worse than the reference execution price.
```

```
    margin_long=100,
    margin_short=100)
```
have capital covering 100% of the position's value.

```
fastLength = input.int(20, "Fast EMA", minval=1)
slowLength = input.int(50, "Slow EMA", minval=1)
```
20: value.
minval=1: prevents values below one.

```
fastEMA = ta.ema(close, fastLength)
slowEMA = ta.ema(close, slowLength)
```
ta = technical analysis.
ema = calculates an exponential moving average.
close = each bar's closing price.
fast Length= the length

```
buySignal = ta.crossover(fastEMA, slowEMA)
```
buySignal = true
fast EMA is > slow EMA.

```
exitSignal = ta.crossunder(fastEMA, slowEMA)
```
exitSignal = True
fast EMA is < slow EMA.

```
if buySignal
   strategy.entry("Long", strategy.long)

if exitSignal
   strategy.close("Long")

plot(fastEMA, title="Fast EMA", color=color.orange)
plot(slowEMA, title="Slow EMA", color=color.blue)
```

3. Results

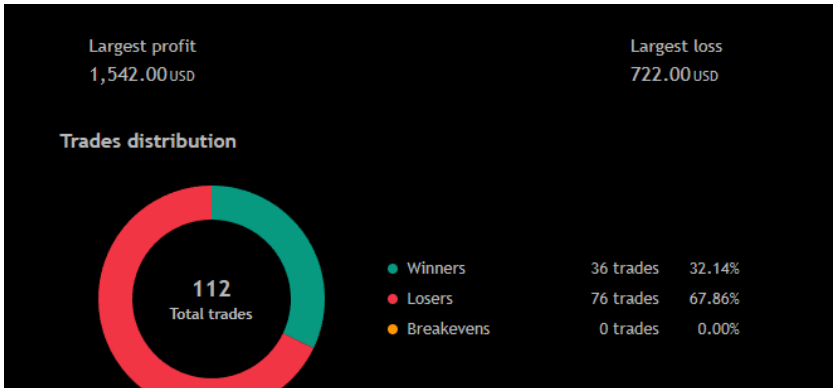


*Figure **Error! No text of specified style in document.**-1simple crossover strategy*

Findings and limitations

During the reported test period, from January 2022 to September 2025, 36 of the 112 closed trades were profitable, resulting in a win rate of 32.14%. The profit factor was 0.938, meaning the strategy generated approximately $0.94 in gross profit for every $1 of gross loss. The strategy was therefore unprofitable over this sample.

Limitations

Long-only approach: The strategy could benefit from rising prices but could not open short positions to profit directly from falling prices.
No protective stop-loss: The strategy used no stop-loss orders. Positions remained open until a downward EMA crossover triggered an exit.

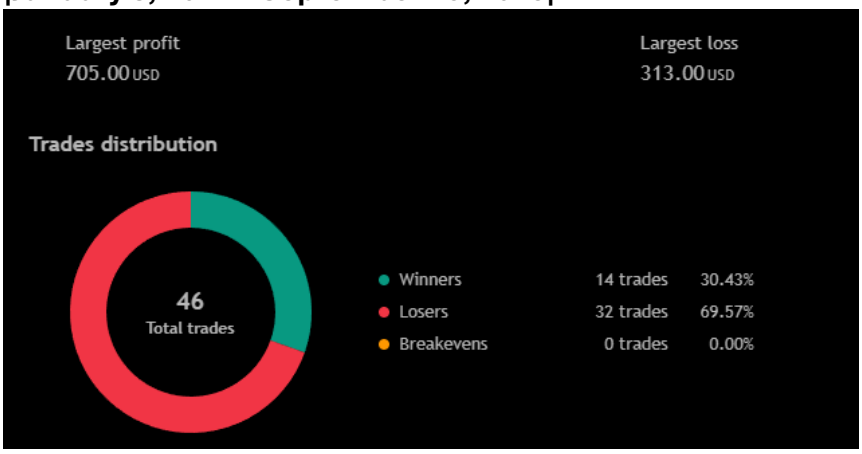
V2| adding a 200 EMA filter

The second version adds a 200-period EMA filter to the baseline strategy. A long position is opened only when the 20-period EMA crosses above the 50-period EMA and the closing price is above the 200-period EMA. The purpose of this modification is to avoid entering long positions when the price is below the longer-term trend indicator.

```
trendEMA = ta.ema(close, 200)
buySignal = ta.crossover(fastEMA, slowEMA) and close > trendEMA
```

Results

|January 3, 2022 – September 18, 2025|



The 200-period EMA filter completed 46 trades, comprising 14 winning trades and 32 losing trades, resulting in a win rate of 30.43%. The largest winning trade generated $705, while the largest losing trade lost $313. The filter reduced trading activity and the largest individual loss in this sample. However, net profit, profit factor, and maximum drawdown are still needed to determine whether it improved overall profitability and risk-adjusted performance.

Findings and limitations

The 200-period EMA filter reduced the number of trades from 112 to 46 and lowered the largest individual loss, while the win rate decreased from 33.93% to 30.43%.

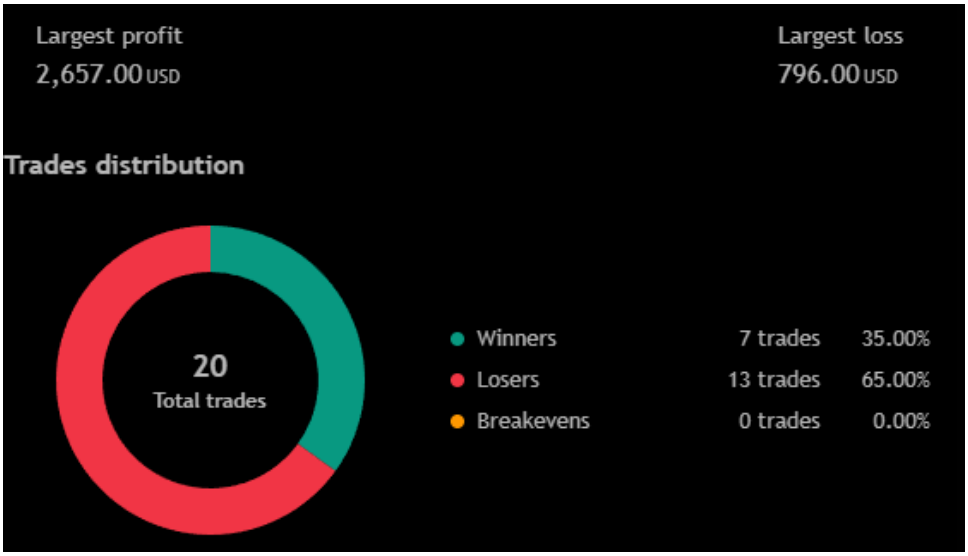
out of sample testing tasting

|OCT1 2025 – 10 NOV 2026|

The modified strategy, including the 200-period EMA filter, was evaluated over a separate historical period. The entry and exit rules, position size, commission, and slippage assumptions were kept unchanged to assess performance beyond the development sample.

Findings and limitations

The strategy completed 20 trades during the evaluation period, comprising seven winning trades and 13 losing trades, resulting in a win rate of 35%. the strategy generated a net profit of $1,706 and a profit factor of 1.524. However, performance was heavily concentrated in the largest winning trade. The small sample of 20 trades limits the strength of the conclusions.



Code with date-filter

```
strategy("Energy Futures — EMA Backtest",
    overlay=true,
    initial_capital=10000,
    currency=currency.USD,
    default_qty_type=strategy.fixed,
    default_qty_value=1,
    pyramiding=0,
    commission_type=strategy.commission.cash_per_contract,
    commission_value=1.50,
    slippage=2,
    margin_long=100,
```

```
    margin_short=100)

fastLength = input.int(20, "Fast EMA", minval=1)
slowLength = input.int(50, "Slow EMA", minval=1)

startDate = input.time(timestamp("01 Oct 2025 00:00 +0000"), "Test start")
endDate = input.time(timestamp("10 Nov 2026 00:00 +0000"), "Test end")

inDateRange = time >= startDate and time < endDate

fastEMA = ta.ema(close, fastLength)
slowEMA = ta.ema(close, slowLength)
trendEMA = ta.ema(close, 200)

buySignal = ta.crossover(fastEMA, slowEMA) and close > trendEMA
exitSignal = ta.crossunder(fastEMA, slowEMA)

if buySignal and inDateRange
    strategy.entry("Long", strategy.long)

if exitSignal
    strategy.close("Long")

if time >= endDate and strategy.position_size > 0
    strategy.close("Long", comment="End of test")

plot(fastEMA, title="Fast EMA", color=color.orange)
plot(slowEMA, title="Slow EMA", color=color.blue)
plot(trendEMA, title="Trend EMA", color=color.red)
```

Key Metrics comparison

Metric	Baseline: 20/50 EMA crossover	V2: 200-period EMA filter	Out-of-sample test
Initial capital	$10,000	$10,000	$10,000
Net profit	−$176.50	−$936.00	+$1,706.00
Return on initial capital	−1.76%	−9.36%	+17.06%
Total closed trades	112	46	20
Winning trades	38	14	7
Losing trades	74	32	13
Win rate	33.93%	30.43%	35.00%
Profit factor	0.984	0.773	1.524

Metric	Baseline: 20/50 EMA crossover	V2: 200-period EMA filter	Out-of-sample test
Expectancy per trade	−$1.58	−$20.35	+$85.30
Gross profit	$10,958.00	$3,188.00	$4,960.00
Gross loss	$11,134.50	$4,124.00	$3,254.00
Average winning trade	$288.37	$227.71	$708.57
Average losing trade	$150.47	$128.88	$250.31
Average win / average loss	1.916	1.767	2.831
Largest winning trade	$1,542.00	$705.00	$2,657.00
Largest losing trade	$722.00	$313.00	$796.00
Sharpe ratio[2]	−0.022	−0.159	0.198
Sortino ratio[2]	−0.039	−0.239	0.615
Commission paid	$337.50	$138.00	$60.00

The baseline EMA crossover strategy produced a small loss of $176.50 (−1.76%) across 112 trades, with a profit factor of 0.984. Adding the 200-period EMA filter reduced trading activity to 46 trades and increased the net loss to $936.00 (−9.36%).

The out-of-sample test generated a profit of $1,706.00 (+17.06%) across 20 trades, with a profit factor of 1.524, although its largest winner exceeded total net profit, indicating substantial dependence on one trade. Overall, the filter did not demonstrate improved profitability, while the small out-of-sample size and differing historical coverage limit firm conclusions.

core performance dashboard

Metric	***Explanation***	***Out-of-sample result***
Profit factor	*Total profit from winning trades divided by total losses from losing trades*	*1.524*
Expectancy per trade	*Average net profit or loss per completed trade*	*+$85.30*
Maximum drawdown	*Largest decline from a previous equity peak*	*$2,017.50 intrabar; reported drawdown of 16.69%*
Sharpe ratio	*Excess return relative to overall return variability*	*0.198*
Sortino ratio	*Excess return relative to downside variability*	*0.615*
Win rate and average win/loss	*Percentage of profitable trades and average winner relative to average loser*	*35% win rate; average winner 2.831 times the average loser*

Metric	***Explanation***	***Out-of-sample result***
Number of trades	*Total number of completed trades in the test*	*20*

Profit Factor

the strategy generated approximately $1.52 in winning-trade profits for every $1 lost on losing trades.

$$Profit\ Factor = \frac{|\ Total\ losses\ from\ losing\ trades\ |}{Total\ Losses\ from\ losing\ trades}$$

Expectancy

The strategy averaged a net profit of approximately $85.30 per completed trade in the out-of-sample test.

Expectancy=P(win)×Average win−P(loss)×Average loss

Sharpe ratio

The Sharpe ratio of 0.198 indicates that the strategy generated modest excess returns relative to the variability of its returns. TradingView uses a default annual risk-free rate of 2%, adjusted to match the monthly return period used in the calculation.

$$\text{Sharpe Ratio} = \frac{MR - RFR}{SD}$$

RFR = 2% per year TradingView's default assumption.

$$SD = \sqrt{\frac{\sum(R_t - MR)^2}{n-1}}$$

R_t = each monthly return
MR= average monthly return
n= number of monthly observations

$$MR = \frac{nR1 + R2 + \cdots + Rn}{n}$$

Sortino Ratio

The Sortino ratio of 0.615 indicates that the strategy earned 0.615 units of excess return per unit of downside deviation.

$$Sortino\ ratio = \frac{MR - RFR}{DD}$$

DD = downside deviation, measuring returns falling below the target return rather than all return fluctuations

Conclusion

This study tested whether a simple EMA crossover strategy could generate consistent returns in Micro WTI futures. The baseline performed close to break-even, while adding a 200-period EMA filter reduced trading activity and drawdown but did not improve profitability. The out-of-sample test produced a positive return of 17.06%, although its small sample of 20 trades and reliance on one large winner limit the strength of this result. Differences in historical coverage also prevent a fully matched comparison between the baseline and filtered versions. Overall, the findings are encouraging but do not establish a robust trading edge.

Overall, this exercise demonstrates how a trading idea can be translated into clear rules, tested against historical data, and evaluated beyond profit alone.